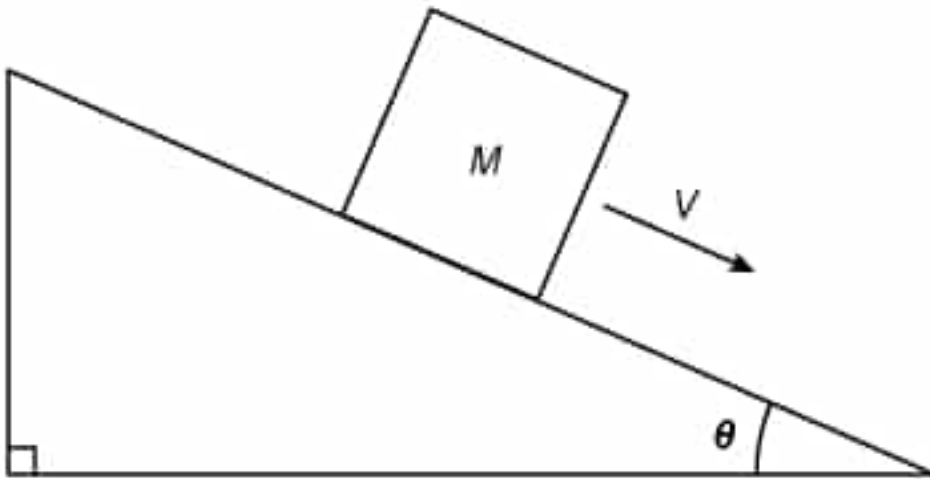


A block with mass M slides down an incline of angle θ at constant velocity v .



Which of the following expressions is equal to the magnitude of the frictional force between the block and the incline? Note: The acceleration of gravity $g = 10 \text{ m/s}^2$.

- ✓ ☒ A. $Mg \sin \theta$ [52%]
☐ B. $Mg \cos \theta$ [37%]
☐ C. $\frac{M \cdot g}{\sin \theta}$ [6%]
☐ D. $\frac{M \cdot g}{\cos \theta}$ [3%]

Correct

52% answered correctly

Explanation:

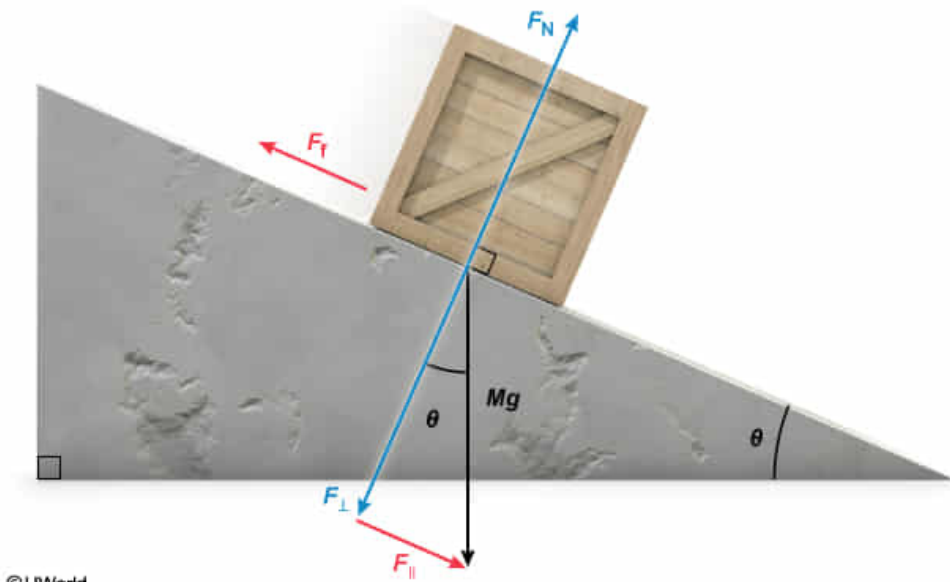
Dynamic equilibrium

No net force

$$F_f = F_{\parallel} = Mg \cdot \sin \theta$$

$$F_{\perp} = F_N$$

- F_f Frictional force
- $F_{\parallel}$ Component of weight parallel to incline
- M Mass
- g Acceleration of gravity
- θ Angle of incline
- $F_{\perp}$ Component of weight perpendicular to incline
- F_N Normal force



According to [Newton's second law of motion](#), an object will accelerate if the net force acting along each axis of a given coordinate system is a nonzero quantity. Conversely, **equilibrium** occurs when the net force acting along each axis of the coordinate system equals zero such that the object does not accelerate.

Dynamic equilibrium describes an object moving at a *constant velocity without acceleration*. When the [components](#) of each force vector are known, [free-body diagrams](#) can be used to assess equilibrium resulting from multiple forces, where:

$$\Sigma_{x \text{ forces}} = 0 = (+x \text{ forces}) + (-x \text{ forces})$$

$$\Sigma_{y \text{ forces}} = 0 = (+y \text{ forces}) + (-y \text{ forces})$$

$$\Sigma_{z \text{ forces}} = 0 = (+z \text{ forces}) + (-z \text{ forces})$$

In this question, the vector components of the forces acting on the block sliding down an incline at constant velocity can be classified as acting either parallel ($\parallel$) or perpendicular ($\perp$) to the surface of the incline:

$$\Sigma_{\parallel \text{ forces}} = 0$$

$$\Sigma_{\perp \text{ forces}} = 0$$

Accordingly, the block experiences a **frictional force** F_f that is equal and opposite to the component of the gravitational force that acts parallel to the surface of the incline $F_{\parallel}$:

$$\Sigma_{\parallel \text{ forces}} = 0 = F_f - F_{\parallel}$$

$$F_f = F_{\parallel}$$

Furthermore, $F_{\parallel}$ can be expressed in terms of the **weight** of the object (Mg) and the angle of incline (θ):

$$F_f = F_{||} = Mg \sin \theta$$

Therefore, the frictional force acting on a mass sliding down an incline in dynamic equilibrium is equal in magnitude to the weight of the object multiplied by the sine of the incline angle **(Choice A)**.

(Choice B) $Mg \cos \theta$ equals the **normal force** exerted on the block by the surface of the incline, not the force parallel to the surface.

(Choices C and D) These equations equal the weight of the object divided by either $\sin \theta$ or $\cos \theta$, which do not represent the parallel or perpendicular components of the force.

Educational objective:

Dynamic equilibrium occurs when the sum of all forces acting on an object in motion is equal to zero (ie, the object moves but does not accelerate).

Free-body diagrams are used to assess the directional components of translational forces.

Time spent:0 sec

Subject:

Physics

QID:402441

System:

Mechanics and Energy